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Microhomology-Mediated Circular DNA Formation from Oligonucleosomal Fragments During Spermatogenesis

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Abstract

The landscape of extrachromosomal circular DNA (eccDNA) during mammalian spermatogenesis, as well as the biogenesis mechanism remains to be explored. Here, we revealed widespread eccDNA formation in human sperms and mouse spermatogenesis. We noted that germline eccDNAs are derived from oligonucleosomal DNA fragmentation in cells likely undergoing cell death. Interestingly, small-sized eccDNAs are associated with euchromatin, while large-sized ones are preferentially generated from heterochromatin. By comparing sperm eccDNAs with meiotic recombination hotspots and structural variations, we found that they are barely associated with *de novo* germline deletions. We further developed a bioinformatics pipeline to achieve nucleotide-resolution eccDNA detection even with the presence of microhomologous sequences that interfere with precise break-point identification. Empowered by our method, we provided strong evidence to show that microhomology-mediated end joining is the major eccDNA biogenesis mechanism. Together, our results shed lights on eccDNA biogenesis mechanism in mammalian germline cells and provide a potential new way for quality assessment of sperms.

eLife assessment

This study provides **valuable** information on the biogenesis of eccDNAs during spermatogenesis, i.e., eccDNAs in spermatogenic cells are not derived from mitotic recombination hotspots but represent oligonucleosomal DNA fragments from apoptotic male germ cells, whose ends are ligated through microhomology-mediated end-joining. The study is currently **incomplete** because the method of bioinformatics needs more details and data interpretation should take the amplification bias into consideration.

Introduction

Apart from linear chromosome, extrachromosomal DNA in circular form also exists in the nuclei of eukaryotes ((Noer *et al.*, 2022)). Circular DNAs could be roughly classified into two groups based on their cell origins ((Chiu *et al.*, 2020)). The one exclusively present in cancerous cells is usually referred to as ecDNA (extrachromosomal DNA), which is of megabase long in average and plays key roles in tumorigenesis ((T. Wang *et al.*, 2021)). The ecDNA biogenesis is linked to ‘episome model’ ((Carroll *et al.*, 1988)), chromothripsis ((Shoshani *et al.*, 2021)), breakage-fusion-bridge ((Coquelle *et al.*, 2002)), or translocation-deletion-amplification model ((Van Roy *et al.*, 2006)). The other class found in somatic and germ cells is usually called eccDNA (extrachromosomal circular DNA). The size of eccDNA ranges from dozens of bases to hundreds of kilobases ((Paulsen *et al.*, 2018)). Contrary to ecDNAs, the biogenesis mechanisms and biological functions of eccDNAs are relatively less experimentally characterized, and current studies show inconclusive or even contradictory results.

The genomic origins of eccDNAs have been extensively investigated in different cells and conditions with the application of Circle-seq and its refined derivatives ((Mehta *et al.*, 2020); (Moller *et al.*, 2015); (Mouakkad-Montoya *et al.*, 2021); (Y. Wang *et al.*, 2021)), where eccDNAs are detected via rolling circle amplification and deep sequencing. While CpG islands ((Shibata *et al.*, 2012)), gene-rich regions ((Moller *et al.*, 2018)), and repeat elements, *e.g.*, LTR (long terminal repeat) ((Moller *et al.*, 2015)), LINE-1 ((Dillon *et al.*, 2015)), segmental duplication ((Mouakkad-Montoya *et al.*, 2021)), or satellite DNA ((Mouakkad-Montoya *et al.*, 2021)), are hotspots for eccDNA formation, others found that eccDNAs are nearly random with regard to genomic distribution ((Moller *et al.*, 2020)), or even made opposite observations ((Henriksen *et al.*, 2022)). Epigenomically, the overall higher GC content, the periodicities of dinucleotide and eccDNA size convergently point to that nucleosome wrapping of DNA might contribute to the formation of small-sized eccDNAs ((Shibata *et al.*, 2012); (Y. Wang *et al.*, 2021)), an intriguing starting point for mechanistic understanding of eccDNA origination. However, direct evidence for coincident positioning of eccDNAs and nucleosomes is still lacking, not to mention specific epigenetic marks on nucleosomes that are tightly associated with eccDNA formation.

EccDNAs are increased upon DNA damages ((Moller *et al.*, 2015); (Paulsen *et al.*, 2021)), suggesting them as byproducts of successive DNA repairs. Among diverse repair pathways, it was reported that eccDNA levels particularly depend on resection after double-strand DNA break (DSB) and repair by microhomology mediated end joining (MMEJ) ((Paulsen *et al.*, 2021); (Y. Wang *et al.*, 2021)). In further support of the involvement of MMEJ, microhomology is found around eccDNA break-points ((Lukaszewicz *et al.*, 2021); (Moller *et al.*, 2015)).

However, in all studies using short-read sequencing technologies, eccDNA break-points are mis-annotated if microhomologous sequences are present around due to their interference to precise break-point detection (see our main text) (([Dillon et al., 2015](#)); ([Henriksen et al., 2022](#)); ([Kumar et al., 2017](#)); ([Lv et al., 2022](#)); ([Mann et al., 2022](#)); ([Moller et al., 2018](#)); ([Moller et al., 2020](#)); ([Paulsen et al., 2019](#)); ([Prada-Luengo et al., 2019](#)); ([Shibata et al., 2012](#)); ([Sin et al., 2020](#)); ([Y. Wang et al., 2021](#)); ([P. Zhang et al., 2021](#))), or the eccDNA identification doesn't depend on precise break-point detections at all (([Moller et al., 2015](#)); ([Mouakkad-Montoya et al., 2021](#))). The contribution of microhomology to eccDNA generations thus needs to be revisited with precise mapping of break-points.

Alternative or additional mechanisms might be involved in germline eccDNA formation. During meiosis, two spatially-closed break sites catalyzed by SPO11 at recombination hotspots may release eccDNAs accompanied by *de novo* deletions on linear chromosomes (([Lukaszewicz et al., 2021](#))). Consistently, germline microdeletions display similar sequence features with eccDNAs (([Shibata et al., 2012](#))). However, a recent study reported that the creation of germline eccDNAs negatively correlate with meiotic recombination rates (([Henriksen et al., 2022](#))). Therefore, it remains to be determined whether meiosis might significantly contribute to eccDNA biogenesis.

We envision that eccDNA landscape during spermatogenesis is ideal for clarifying the abovementioned issues and so better understand the biogenesis mechanisms and biological implications of eccDNAs. Only a small fraction of histones will survive from the histone-to-protamine transition in mature sperms, allowing us to more specifically correlate eccDNA origination with histones. Studying eccDNAs in germline cells rather than somatic cells could help reveal to what extent meiosis might contribute to eccDNA generation and *de novo* structural variations that can be passed to offspring. Therefore, in this study, we profiled eccDNAs via Circle-seq in human sperms and different developmental stages of mouse germ cells with an improved analysis pipeline to identify eccDNAs at nucleotide resolution. We conclude that germline eccDNAs are formed largely by microhomology mediated ligation of nucleosome-protected fragments, and barely contribute to *de novo* genomic deletions at meiotic recombination hotspots.

Results

Widespread eccDNA formation in human and mouse germline cells

Because it was reported that high content of eccDNAs existed in sperms (([Henriksen et al., 2022](#))), we examined the genome-wide eccDNA landscape in two human sperm samples by Circle-seq (([Moller et al., 2018](#)); ([Moller et al., 2015](#)); ([Moller et al., 2020](#))) (see Materials and methods), and indeed found that there were widespread eccDNAs across the human genome ([Figure 1A](#), [Figure 1—figure supplement 1A](#)). This motivated us to further investigate the biogenesis mechanism, particularly whether it might be linked to specific spermatogenesis processes. Given that it is ethically prohibited and technically challenging to collect pure spermatogenic cell types from human individuals, we therefore turned to use mouse model to study the eccDNA formation during spermatogenesis.

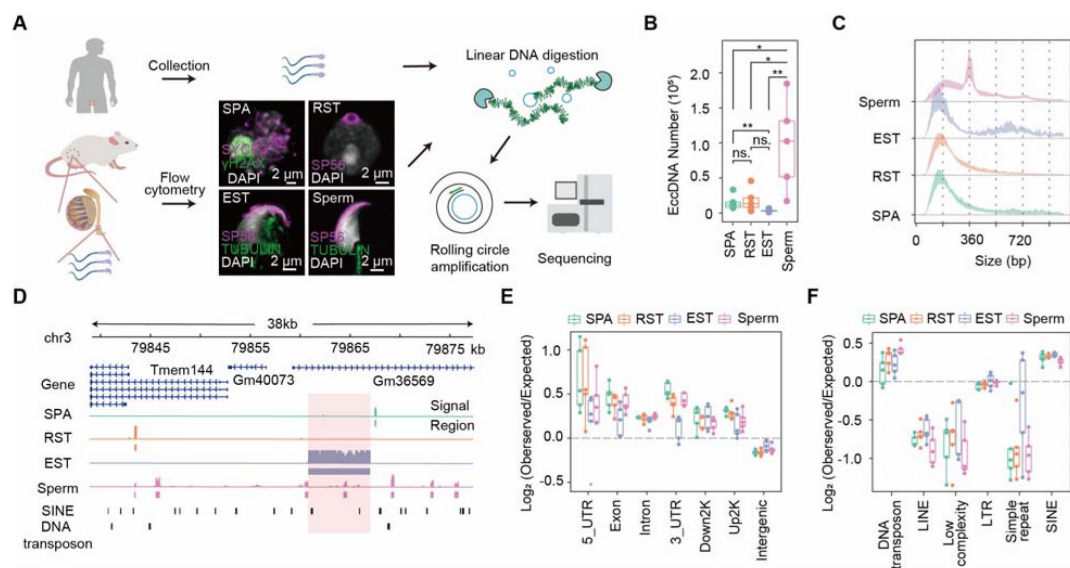


Figure 1.

Overview of eccDNA formation during mouse spermatogenesis. **(A)** Schematic representation of Circle-seq in human sperm cells and mouse SPA, RST, EST and sperm cells validated with immunochemistry. SYCP3: a component of the synaptonemal complex; γ H2AX: a marker for double-strand breaks; SP65: a marker for acrosome organelle; TUBULIN: structural component of manchette in EST and flagellum axoneme in sperm cells. **(B)** Number of eccDNAs detected in different cell types. *two-sided t -test p -value < 0.05 ; ** two-sided t -test p -value < 0.01 . **(C)** Size distribution of eccDNAs during mouse spermatogenesis. Dotted lines indicate multiples of 180 bp. **(D)** A representative genomic locus showing the gene annotation, Circle-seq signals, detected eccDNAs, and SINE and DNA repeat elements. Highlighted in red rectangle is a large-sized eccDNA. **(E)** Enrichment of eccDNAs at given genomic regions relative to randomly-selected control regions. **(F)** Enrichment of eccDNAs at given repeat elements relative to randomly-selected control regions.

A series of cell divisions and morphological changes are involved in spermatogenesis, where spermatogonial stem cells develop into spermatocytes (SPA) via mitosis, and SPA then undergo meiosis to produce haploid round spermatids (RST), which will take a dramatic morphological change and chromatin compaction to produce elongated spermatids (EST) and finally matured sperms ((Hess et al., 2008); (Roosen-Runge, 1962)). We isolated SPA, RST and EST using flow cytometry and collected sperms from mouse cauda epididymis ((Hayama et al., 2016)) for subsequent Circle-seq ((Moller et al., 2018); (Moller et al., 2015); (Moller et al., 2020)) (see Materials and methods). All four cell types were validated with known markers and cell morphology (Figure 1A). To account for sample variations, up to five biological replicates and ~150 million reads for each cell type were sequenced for eccDNA detection. From ~1,500 to ~180,000 high-confidence eccDNAs were identified, suggesting widespread circular DNA formation during mouse spermatogenesis (Figure 1B; see Materials and methods). The reproducible rate of eccDNAs with 50% reciprocal overlap between biological replicates was only ~2.4% in average, a level comparable to previous studies ((Henriksen et al., 2022); (Moller et al., 2018)) (Figure 1—figure supplement 1B). As noted earlier ((Moller et al., 2018)), the detected eccDNAs seemed not saturated (Figure 1—figure supplement 1C; see Discussion and Materials and methods), which might underlie the observed low reproducibility. Nevertheless, principal component analysis suggested that the within-group

similarity was marginally higher than the between-group similarity (Figure 1—figure supplement 1D), allowing investigation of stage-specific eccDNA features during mouse spermatogenesis.

The detected germline eccDNAs verified known genomic features of eccDNAs. First, the natural size distribution of eccDNA is usually distorted in Circle-seq as smaller eccDNAs tend to be overrepresented in rolling circle amplification ((Mohsen *et al.*, 2016); (Moller *et al.*, 2018); (Moller *et al.*, 2015)). As expected, the detected eccDNA population was dominated by small-sized eccDNAs, most of which were ~180bp or ~360bp long (Figure 1C). However, eccDNA size could occasionally reach to tens of kilobases (Figure 1D). Second, eccDNAs from different cell types were all enriched at gene-rich regions, especially 5'UTR (Figure 1E, Figure 1—figure supplement 1E), corroborating the reported association between eccDNA frequency and gene density in somatic cells ((Dillon *et al.*, 2015); (Shibata *et al.*, 2012)). EccDNAs were also highly associated with SINE but not LINE elements (Figure 1F), and quantitative analysis revealed that eccDNA biogenesis was positively correlated with SINE density (Figure 1—figure supplement 1F), but negatively correlated with LINE density (Figure 1—figure supplement 1G). Given that SINE and LINE elements function to orchestrate chromosomes into gene-rich A compartment and gene-poor B compartment respectively ((Lu *et al.*, 2021)), the positive correlation between eccDNAs and SINE elements might further support that eccDNAs are overall highly associated with the gene-rich regions. Interestingly, we also noticed a strong association between eccDNAs and DNA transposons (Figure 1F), suggesting that DNA transposons might get circularized rather than or in addition to re-integrated into the genome, an interesting possibility awaiting further investigations. Altogether, the genome-wide eccDNA landscape during mouse spermatogenesis allow us to further study the biogenesis mechanism and function of eccDNAs.

High eccDNA load and periodic eccDNA size distribution in mouse sperm cells

Notably, sperm cells had 97,372 eccDNAs detected in average, a number significantly higher than those in SPA (15,246), RST (18,426) and EST (3,591) cells (Figure 1B). SPA cells did not show higher eccDNA numbers (Figure 1B), suggesting that meiosis does not seem to contribute significantly to eccDNA biogenesis. Since the same amount of eccDNAs (10ng) were used for library construction and all samples were sequenced in comparable and sufficiently-deep depth, it suggests that eccDNA species in sperm cells has higher complexity. However, the higher starting cell number for sperm cells might account for the larger diversity of sperm eccDNA species (see Discussion and Materials and methods), otherwise, it would be interesting to explore any specific features of sperm cells underlying the higher load of eccDNAs.

In contrast to SPA, EST and RST eccDNAs showing the unimodal distribution that was centered at ~180bp, sperm-derived eccDNAs showed a multimodal distribution with a pronounced periodicity of ~180bp (Figure 1C), which was readily seen in individual samples (Figure 1—figure supplement 2). Given that each nucleosome consists of 147bp DNA wrapping itself around a histone core, the ~180bp-long fragments likely corresponded to histone core region plus ~20-30bp linker regions, as observed in apoptotic cells ((Matassov *et al.*, 2004)). Although the identified eccDNAs in all spermatogenesis stages were likely related to nucleosomes, the different modes of size distribution might be due to distinct nucleosome compositions and structures between sperm and other spermatogenic cells.

Mouse sperm eccDNAs come from DNA fragments protected by histones

Only a small fraction of histones will be retained in mouse sperm cells after histone-to-protamine transition (*Torres-Flores et al., 2020*), permitting us to more specifically correlate eccDNAs with histones. We were therefore motivated to see whether the detected eccDNAs were derived from the retained histones in mature sperm cells. We noted that sperm eccDNAs had higher GC content than surrounding regions as well as control regions randomly selected across the genome (*Figure 2A*), resembling the sequence feature of nucleosome-protected DNA fragments. Consistently, sequence-based prediction revealed significantly higher nucleosome occupancy probability for ~180 bp (from 175bp to 185bp) and ~360bp (from 355bp to 365bp) sperm eccDNA regions (*Figure 2B*; see Materials and methods). A small dip was observed at the center of ~360bp eccDNA regions, which likely corresponded to the linker region between two nucleosomes (*Figure 2B*, right).

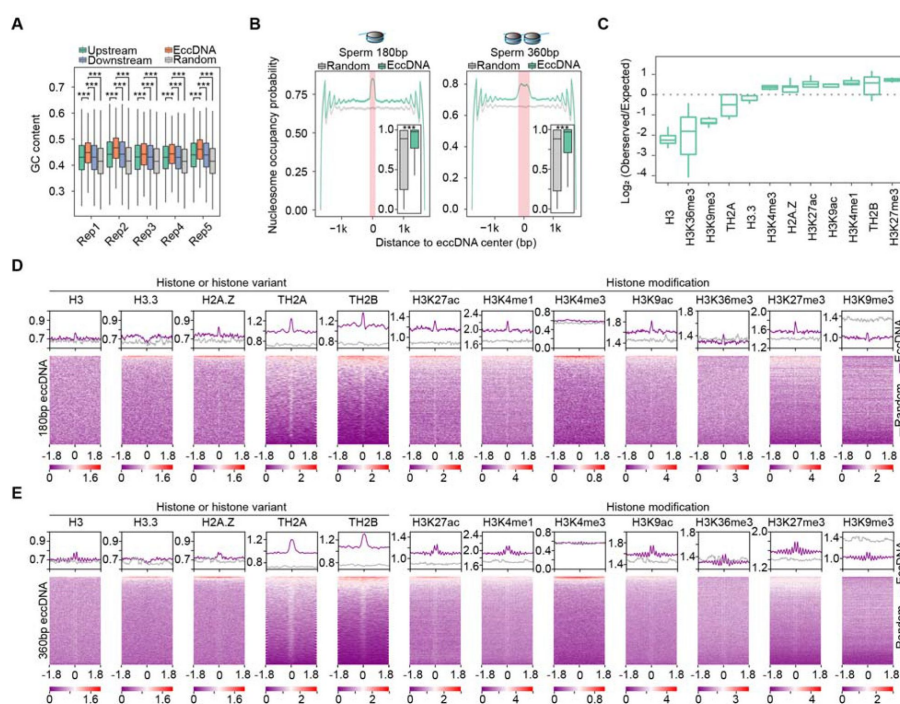


Figure 2.

Association between sperm eccDNAs and nucleosome positioning. (A) GC contents of sperm eccDNAs, regions upstream and downstream of eccDNAs, and randomly-selected length-matched control regions. ***two-sided *Wilcoxon* test *p*-value <0.001. (B) Predicted probability of nucleosome occupancy for eccDNA and randomly-selected length-matched control regions (highlighted by red-shaded area), and surrounding regions. Boxplots showing the probability distribution of individual eccDNAs and control regions. ***two-sided *Wilcoxon* test *p*-value <0.001. (C) Enrichment of eccDNAs at different histones and histone modifications. (D and E) ChIP-seq signal distribution at [-1.8kb, +1.8kb] of ~180bp (D) and ~360 bp (E) eccDNA regions. ChIP-seq signals quantified as reads density are color-coded below heatmaps.

It is a common practice to re-use publicly-available genomics data generated in the same cell types for integrative analysis. Taking advantage of public ChIP-seq data for histones and their modifications in mouse sperm cells (*Jung et al., 2019*; *Jung et al., 2017*; *Singh et al., 2021*), we found that eccDNAs were significantly enriched with certain histone variants and modifications (*Figure 2C*), and 7.46% of sperm eccDNAs in total were intersected with at least one ChIP-seq peaks. Considering that histones occasionally retained in sperms might not generate strong ChIP-seq signals exceeding the peak calling cutoff, a meta-gene analysis of ChIP-seq signals at and around sperm eccDNA regions will likely provide more insights. Interestingly, enrichment of H3 histone and H2A.Z, TH2A and TH2B histone variants but depletion of H3.3 variant was observed at ~180bp sperm eccDNA regions (*Figure 2D*). These

eccDNAs also showed strong associations with H3K27ac, H3K4me1, H3K9ac and H3K27me3 modifications, however, no enrichment was seen for H3K4me3, and H3K36me3 and H3K9me3 signals were comparable with or even lower than randomly-selected regions as control (Figure 2D).

We next examined ~360bp sperm eccDNAs which supposedly correspond to two nucleosomes, and made similar observations. Centers of ~360bp eccDNAs were well positioned between two adjacent nucleosomes consisting H3 histone and H2A.Z histone variants, and H3K27ac, H3K4me1, H3K9ac and H3K27me3 histone modifications (Figure 2E). Similar to ~180bp eccDNAs, ~360bp eccDNAs did not show association with H3.3 or H3K4me3, or stronger association than randomly-selected regions with H3K36me3 and H3K9me3 either (Figure 2E). Although H3.3 variant coincides with active transcription, it is also well known for its localization at heterochromatin region and its roles in promoting heterochromatin formation by inhibiting H3K9/K36 histone demethylase (*Udugama et al., 2022*). Together, euchromatin is generally more preferred than heterochromatin for eccDNA biogenesis, which is consistent with the enrichment of sperm eccDNAs at gene-rich regions (Figure 1D).

Large-sized eccDNAs are preferentially generated from heterochromatin regions

Intriguingly, periodic distribution of nucleosomes was observed for ~360bp eccDNAs but not for ~180bp ones, indicating that eccDNAs from di-nucleosomes but not mono-nucleosomes preferentially originate from well-positioned nucleosome arrays (Figure 2E). We were further prompted to ask whether eccDNAs of different sizes are originated from different genomic regions. Indeed, small-sized eccDNAs (<3kb) were more enriched at H3K27ac-marked euchromatin regions, while large-sized ones (≥3kb) at H3K9me3-marked heterochromatin regions (Figure 3A). Accordingly, small-sized eccDNAs were generally more associated with genic regions, while large-sized ones with non-genic regions (Figure 3B). Since sperm eccDNAs in this study were dominantly small-sized ones (Figure 1C), strong enrichment of eccDNAs at genic regions was observed (Figure 1E). However, strong depletion at genic regions was reported for human sperm eccDNAs in a recent study (*Henriksen et al., 2022*). Close inspection suggests that the discrepancy is partially reconciled in the light of two eccDNA groups of different sizes. Henriksen *et al* studied eccDNAs with the size largely ranging from ~3kb to 50kb (*Henriksen et al., 2022*), rather than small-sized ones reported by us and many others (*Dillon et al., 2015*; *Moller et al., 2018*; *Moller et al., 2020*; *Paulsen et al., 2019*; *Shibata et al., 2012*; *Y. Wang et al., 2021*). In contrast to small-sized eccDNAs, the large-sized sperm eccDNAs detected in this study also displayed weak but apparent negative correlation with gene density and Alu elements (Figure 3C and D).

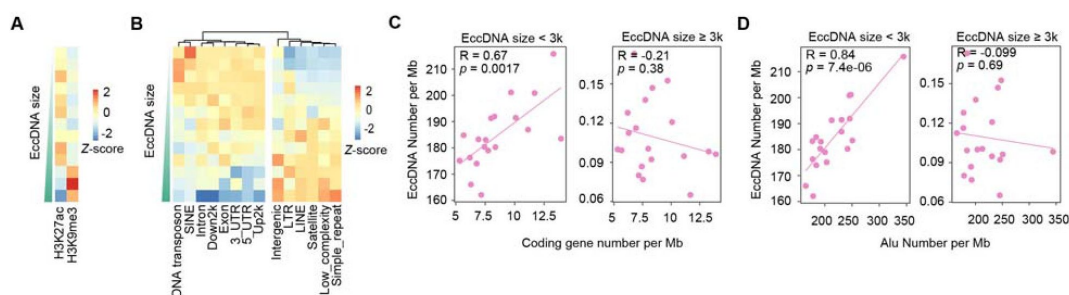


Figure 3.

Large-sized eccDNAs are preferentially generated from heterochromatin regions. (A) Distribution at H3K27ac- and H3K9me3-marked regions for eccDNAs of different sizes. (B) Distribution at different genomic regions for eccDNAs of different sizes. (C) Number of small (<3kb) vs large (≥3kb) eccDNAs per Mb as a function of gene number per Mb. *Pearson* correlation coefficients and two-sided *t*-test *p*-values are indicated. (D) Number of small (<3kb) vs large (≥3kb) eccDNAs per Mb as a function of Alu number per Mb. *Pearson* correlation coefficients and two-sided *t*-test *p*-values are indicated.

Germline eccDNAs as apoptotic products are not associated with meiotic recombination hotspots

The observed association between eccDNA and oligonucleosomal DNA fragmentation (Figure 2) is a typical feature of cell death. The spontaneous death of germ cells has been observed during the normal spermatogenesis (Liu et al., 2017; Shaha et al., 2010; Weinbauer et al., 2001; Young et al., 2001), however, it is still debatable whether spermatids and sperm can undergo apoptosis (Lachaud et al., 2004). Thus, sperm-derived eccDNAs might be associated with apoptosis (if exists) or un-programed cell death of germ cells during the normal spermatogenesis (see also Discussion). In support of this hypothesis, all features associated with mouse germline eccDNAs identified in this study (Figure 1C, E and F) closely matched with those of eccDNAs whose generation is dependent on apoptotic DNA fragmentation in somatic cells (Figure 4—figure supplement 1) (Y. Wang et al., 2021).

During meiosis, two spatially-closed cleavage sites catalyzed by SPO11 at recombination hotspots could release eccDNAs and generate *de novo* genomic deletions (Lukaszewicz et al., 2021), which, if transmitted to offspring, might contribute to structural variations within population. Since most sperm eccDNAs likely result from oligonucleosomal DNA fragments in sperm cells undergoing cell death (Figures 2 and 3) and SPA cells undergoing meiosis does give rise to more eccDNAs than other cells (Figure 1B), meiotic recombination is unlikely the major mechanism for germline eccDNA generation. To test this hypothesis, we first investigated to what extent eccDNA break-points well correspond to recombination hotspots defined as SPO11 or PRDM9 binding sites (Alleva et al., 2021; Lange et al., 2016). We noted that there was only a small number of eccDNAs with both break-points located in one recombination hotspot or two different hotspots (Figure 4A). These eccDNAs only constituted <0.15% (or <350) of mouse germline eccDNAs, suggesting a very low level of coincidence between eccDNA generation and meiotic recombination (Figure 4A). Consistently, only dozens of, or a few hundred eccDNAs in mouse germline cells coincided with known genomic deletions within mouse population (Figure 4B). Altogether, germline eccDNAs are

likely apoptotic products that are not associated with meiotic recombination hotspots and heritable genomic deletions.

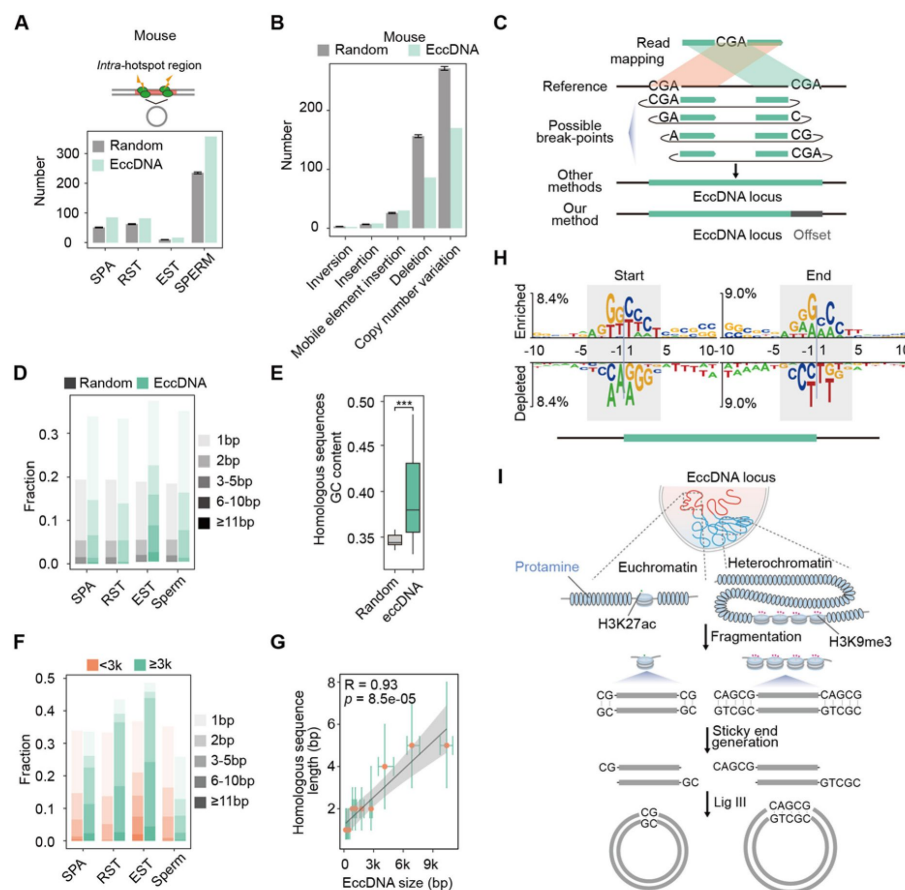


Figure 4.

Microhomology directed ligation accounts for emergence of most eccDNAs. (A) Numbers of eccDNAs or randomly-selected control regions overlapped with recombination hotspots in mouse. (B) Shown are numbers of mouse sperm eccDNAs or randomly-selected control regions having 95% reciprocal overlap with different types of structural variations. (C) Illustrated are how an eccDNA with homologous sequences (CGA) at two ends is identified from short-read sequencing data by our methods vs other methods. (D) Percentages of homologous sequences of different lengths (coded by different color saturation levels) are shown for eccDNAs and randomly-selected control regions. (E) GC content of homologous sequences and randomly-selected control regions. (F) Percentages of homologous sequences of different lengths (coded by different color saturation levels)

are shown for small-sized (<3kb) and large-sized (≥3kb) eccDNAs. (G) Length of microhomologous sequences as a function of the eccDNA size. Data points are shown as median plus lower (25%) and upper (75%) quartiles. The shaded area is 95% confidence interval of linear regression line. *Pearson* correlation coefficient and two-sided t-test *p*-value are indicated. (H) Sequencing motif analysis for ±10bp leftmost left ends and ±10bp leftmost right ends of eccDNAs with no perfectly matched homologous sequences observed. (I) Model for MMEJ-directed eccDNA biogenesis.

Microhomology directed ligation is the major biogenesis mechanism of germline eccDNAs

We therefore further explored how nucleosome-protected DNA fragments get circularized into eccDNAs. As suggested by previous studies, MMEJ is implicated in eccDNA biogenesis ((Lukaszewicz et al., 2021); (Moller et al., 2015); (Paulsen et al., 2021); (Y. Wang et al., 2021)). The precise distribution of microhomologous sequences relative to eccDNA break-points will help better understand how and to what extent MMEJ might contribute to eccDNA biogenesis. However, we noted that the presence of microhomologous sequences will hinder precision eccDNA break-point identification (Figure 4C), which is not well dealt with by all existing methods (Figure 4—figure supplement 2A) ((Kumar et al., 2017); (Mann et al., 2022); (Prada-Luengo et al., 2019); (P. Zhang et al., 2021)). Short sequencing reads spanning eccDNA break-points will be mapped to the genome as split reads, with its first part mapped to the

right end of eccDNA, and the second part to the left end. If the sequence in front of the left eccDNA end is homologous to the right eccDNA end, or if the sequence following the right eccDNA end is homologous to the left eccDNA end, the homologous regions will be included in both parts of split reads to reach to a maximal length of matches, and many existing methods will mistake the eccDNA plus two homologous regions as the whole eccDNA region (Figure 4C). Being aware of it, we developed a base-resolution method for eccDNA identification on the basis of previous efforts (Figure 4—figure supplement 2B) ((Kumar *et al.*, 2017); (Moller *et al.*, 2018)). When homologous sequences are present, we record the coordinates of the leftmost form of eccDNA and an offset corresponding to the length of homologous sequences to represent all possible eccDNA variants (Figure 4C). Our method has comparable sensitivity and specificity with existing methods (Figure 4—figure supplement 2C), and is easy-to-use and freely-available.

In contrast to simulated controls (15%), more than one third of eccDNAs had ≥ 1 bp homologous sequences, most of which were shorter than 5bp (Figure 4D), suggesting the involvement of MMEJ in eccDNA biogenesis. The GC content of homologous sequences was higher than that of simulated control regions, permitting stronger base-pairing for efficient MMEJ (Figure 4E). Considering that two free-ends of long DNA fragments might be not as spatially close as those of short DNA fragments, formation of longer eccDNA should more rely on longer homologous sequences for stable base pairing. Indeed, large-sized eccDNAs in SPA, RST and EST cells did show higher percentage of ≥ 2 bp homology than small-sized eccDNAs, and large-sized eccDNAs in sperm cells showed higher percentage of > 5 bp homology (Figure 4F). A significant positive correlation between lengths of homologous sequences and eccDNA sizes was observed (Figure 4G). We further reasoned that for the remaining two thirds of eccDNAs that were lack of perfectly-matched homologous sequences, imperfect homologous sequences might be present. Accordingly, we noted the same sequence motifs between eccDNA starts and sequences following eccDNA ends, and between eccDNA ends and sequences in front of eccDNA starts (Figure 4H). Similar observations have been made also by others before ((Sin *et al.*, 2020)), however they failed to precisely locate the homologous sequences relative to eccDNA break-points. Altogether, we propose that sticky ends of DNA fragments with homologous sequences might base pair with each other, and then be ligated by DNA ligase, *e.g.*, DNA ligase III ((Y. Wang *et al.*, 2021)), to form eccDNAs (Figure 4I). In sum, among all proposed mechanisms ((Chiu *et al.*, 2020); (Dillon *et al.*, 2015); (Moller *et al.*, 2015); (Paulsen *et al.*, 2021); (Sin *et al.*, 2020)), MMEJ-mediated ligation accounts for emergence of most eccDNAs at least in germline cells.

The biogenesis mechanism of germline eccDNAs is conserved between human and mouse

Upon revealing the major biogenesis mechanism of mouse germline eccDNAs, we further examined whether the mechanism is conserved in human by analyzing the features of human sperm eccDNAs detected in this study (see Materials and methods). Similarly, human sperm eccDNAs were originated from oligonucleosomal fragmentation as well, as suggested by the pronounced size periodicity of ~ 180 bp (Figure 5A), higher GC content and nucleosome occupancy probability (Figure 5B and C). We further performed base-level comparison between eccDNAs and meiotic recombination hotspots. Consistent with what we observed in mouse, only about 600 eccDNAs ($< 0.7\%$ of human sperm eccDNAs) were located in one recombination hotspot or two different hotspots (Figure 5D), and only dozens of, or a few hundred eccDNAs in human sperm cells coincided with known genomic deletions (Figure 5E). Overall, our analysis disfavors the meiotic recombination to eccDNA biogenesis. Instead, we observed higher frequency of micro-homologous sequences for eccDNAs than simulated control regions (Figure 5F). Moreover, large-sized eccDNAs had longer homologous sequences possibly for stable base pairing, as indicated by the strong positive correlation

between lengths of homologous sequences and eccDNA sizes (Figure 5G). Altogether, these results suggest microhomology directed ligation of nucleosome protected DNA fragments is a conserved pathway for germline eccDNA generation in both human and mouse.

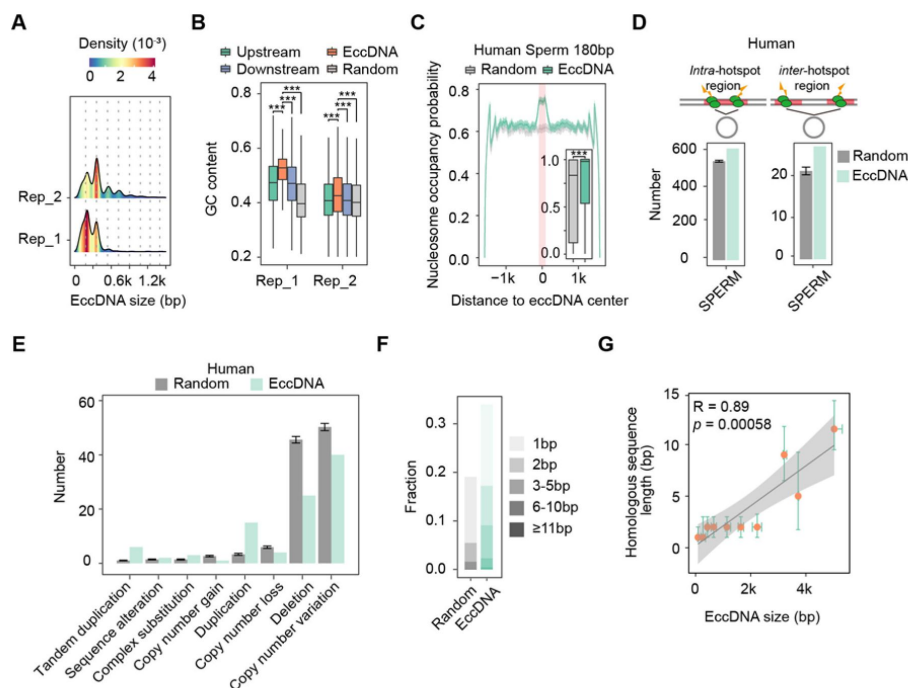


Figure 5.

The biogenesis mechanism of germline eccDNAs is conserved between human and mouse. (A) Size distribution of sperm eccDNAs in two biological replicates. (B) GC contents of sperm eccDNAs, regions upstream and downstream of eccDNAs, and randomly-selected length-matched control regions. ***two-sided Wilcoxon test p -value <0.001 . (C) Predicted probability of nucleosome occupancy for eccDNA and randomly-selected length-matched control regions (highlighted by red-shaded area), and surrounding regions. Boxplots showing the probability distribution of individual eccDNAs and control regions. ***two-sided Wilcoxon test p -value <0.001 . (D)

Numbers of eccDNAs or randomly-selected control regions overlapped with recombination hotspots in human. eccDNAs located completely within a hotspot (*Intra*-), or with both ends overlapped with two different hotspots (*Inter*-) are shown separately. (E) Shown are numbers of human sperm eccDNAs or randomly-selected control regions having 95% reciprocal overlap with different types of structural variations. (F) Percentages of homologous sequences of different lengths (coded by different color saturation levels) are shown for eccDNAs and randomly-selected control regions. (G) Length of microhomologous sequences as a function of the eccDNA size. Data points are shown as median plus lower (25%) and upper (75%) quartiles. The shaded area is 95% confidence interval of linear regression line. Pearson correlation coefficient and two-sided t -test p -value are indicated.

Discussion

The biogenesis mechanisms and biological implications of eccDNAs are big puzzles. Increasingly evidence has emerged to relate eccDNA formation with nucleosome-protected DNA fragments, however, to our knowledge, no study provides direct evidences. The comparison between high number of sperm eccDNAs and retained histones in sperm cells after histone-to-protamine transition ((Torres-Flores *et al.*, 2020)) provide a good way to clarify the abovementioned issues. Here, we profiled eccDNA landscape during mouse spermatogenesis, and explicitly linked eccDNAs with nucleosome-protected DNA fragments (Figure 2). According to these results, we speculate that any oligonucleosomal fragments might have a chance of eccDNA formation as long as microhomologous sequences are present at the ends. In other words, any single cells might generate distinct sets of eccDNAs and thus the theoretical number of eccDNA is extremely large. This might explain why sperm cells, which have a higher starting cell number, have more eccDNAs detected, even

though the same amount of eccDNAs (10 ng) for each cell type was used for library preparation and high-coverage sequencing (~150 million reads).

Multiple biogenesis mechanisms of eccDNA have been proposed ((*Chiu et al., 2020*); (*Dillon et al., 2015*); (*Moller et al., 2015*); (*Paulsen et al., 2021*); (*Sin et al., 2020*)), it is however unclear which one plays the dominant role. With our nucleotide-resolution eccDNA break-point detection method, we clearly demonstrated that microhomologous sequences are present at boundaries of most eccDNAs (Figure 4), strongly suggesting the MMEJ-mediated eccDNA formation as the major mechanism for eccDNA biogenesis. The biogenesis mechanism of germline eccDNAs is conserved between human and mouse. However, it remains unclear whether eccDNA generation from healthy cells not undergoing apoptosis is also mediated by MMEJ.

Spontaneous germ cell death has been observed during the normal spermatogenesis ((*Liu et al., 2017*); (*Shaha et al., 2010*); (*Weinbauer et al., 2001*); (*Young et al., 2001*)). Given the periodic size distribution (Figure 1C) and similar genomic features with eccDNAs from apoptotic DNA fragmentation in somatic cells (Figure 1 and Figure 4—figure supplement 1), we reason that mouse germline eccDNAs may be also the germ cell death products. As the final place for sperm maturation and storage, epididymis might contain unhealthy sperm cells undergoing cell death and so DNA fragments, which might additionally contribute to the high eccDNA loads of sperm cells than others. It is possible that failure of histone-to-protamine exchange might account for some sperm cell death. It has been shown that H3.3, generally linked to H3K4me3 marks, plays key roles in modulating TP1 removal and PRM1 incorporation for nucleosome eviction and replacement by protamine ((*Wang et al., 2019*)), and loss of H3.3 will increase the cell death rate ((*Yuen et al., 2014*)). H3K36me3 and H3K9me3, whose demethylase is inhibited by H3.3 ((*Udugama et al., 2022*)), are also involved in histone replacement ((*Wang et al., 2019*)), failure of which might underlie the eccDNA generation as well. In line with it, sperm eccDNA regions correspond to depletion of H3.3 variant and H3K4me3, H3K36me3 and H3K9me3 histone modifications (Figure 2D and E). This suggests the number of sperm eccDNAs might be highly associated with cell death, and sperm eccDNA may serve as a clinical biomarker for quality assessment of healthy sperms.

It is still debatable whether meiotic recombination is an important source of eccDNA generation. Both positive and negative correlations between eccDNAs and meiotic recombination rates at chromosomal level have been reported previously ((*Henriksen et al., 2022*); (*Lukaszewicz et al., 2021*)). Our results suggest that this discrepancy can be largely reconciled in the light of two eccDNA groups of different sizes. We noted that small-sized and large-sized eccDNAs are preferentially derived from euchromatin and heterochromatins respectively (Figure 3). Since meiotic recombination hotspots are enriched at euchromatin regions ((*Lange et al., 2016*)), small-sized eccDNAs are therefore positively correlated with recombination rates but the opposite is true for large-sized eccDNAs (Figure 3—figure supplement 1). Therefore, the observed correlations between eccDNA density and recombination meiotic rate at chromosomal level are simply indirect reflections of their preferences to different chromatin regions.

We further investigated the coincidence between eccDNAs and meiotic recombination hotspots and structural variations at base level, and found that sperm cell eccDNAs does seem to be a major source or by-products of *de novo* deletions, likely because that extensive eccDNA formation might only occur at dead sperm cells that having no chance to be transmitted to the next generation. However, we cannot fully exclude the possibility that some eccDNAs emerging in normal sperm cells and do not affect their viability will be accompanied by *de novo* deletions or re-integrated into genome to create structural variations. Furthermore, they might stimulate innate immune responses ((*Y. Wang et al., 2021*)), server as extracellular vesicles encoding small RNAs ((*Paulsen et al., 2019*)), and be taken as biomarkers ((*Lv et al., 2022*)), which represent exiting research directions.

Methods

Isolation of mouse germ cells

Testes from adult C57BL/6 mice were decapsulated, and the seminiferous tubules were torn into small pieces. After incubating in 8 ml PBS containing 1 mg/ml collagenase (Sigma, C5138, St. Louis, MO, USA) and 1 mg/ml hyaluronidase (Sigma, H3506, St. Louis, MO) at 37°C for 6 min, the dispersed seminiferous tubules and cells were incubated at 37°C for 5 min with gentle shaking. Cells were collected by centrifugation at $200 \times g$ for 5 min at 4°C, and washed once with PBS, resuspended in 15 ml PBS containing 0.25% Trypsin (Gibco, 25200072) and 1 mg/ml DNase I (Sigma, AMPD1), and incubated at 37°C for 5 min with gentle shaking. Thereafter, cells were collected and washed once with PBS. After filtration through a 70 μm Nylon Cell Strainer (BD Falcon, 352350), the cells were suspended in 30 mL of PBS and stained at 37°C with 5 mg/mL Hoechst 33342 (Thermo Scientific, 62249) for 30 min. Hoechst fluorescence were detected with a 450-nm band-pass filter for blue fluorescent (Hoechst Blue) or a 675-nm band-pass filter for red fluorescent (Hoechst Red). Mouse spermatocytes, round spermatids and elongating spermatids were sorted by BD FACS Aria™ Fusion (BD Biosciences). Mouse mature spermatozoa were collected from mouse cauda epididymis. The cauda epididymis was quickly cut into pieces and incubated in 1 ml pre-warmed human tubal fluid (HTF) (Millipore, MR-070-D) for 15 min at 37°C, thus allowing the mature spermatozoa to release from the tissue. After filtration through a 40 μm Nylon Cell Strainer (BD Falcon, 352340), mature spermatozoa were collected and washed three times with PBS. About 10^6 SPA, RST and EST cells and 10^7 sperm cells were obtained and used for the eccDNA extraction.

Immunofluorescence

Mouse germ cells were spread on glass slides that were air dried. The slides were fixed in 4% PFA at room temperature for 5 min, washed with PBS three times. After blocking with 5% bovine serum albumin, the primary antibodies (rabbit anti-SYCP3 pAb (Abcam, ab15093, 1:400); mouse anti- γ H2AX pAb (Millipore, 05-636, 1:400); Mouse anti-sp56 mAb (QED Bioscience, 55101, 1:200); Rabbit Tubulin pAb (ABclonal, AC007, 1:100)) were added to the slides, followed by 14-16 h incubation at 4°C. After washing the slides with PBS, the secondary antibodies were added, followed by 1.5 h incubation. The cell nuclei were stained with DAPI for 5 min. Images were observed using a fluorescence microscope SP8 microscope (Leica).

Human adult sperm sample preparation

The sperm donation candidates in the present study were healthy young Chinese men. Each candidate completed a medical examination and extensive medical/social questionnaire to exclude any potential individuals with genetic or major medical problems (such as cardiovascular diseases and sexually transmitted diseases) listed in the Basic Standard and Technical Norms of Human Sperm Bank published by Chinese Ministry of Health. Smokers, drug abusers, and heavy drinkers were also excluded. The rest of the candidates signed a voluntary sperm donation informed consent and agreed to live in Beijing for at least 6 months. The sperm bank also recorded the candidates' age, date of birth, and date of semen collection. The ethical approval in this study were provided by the Reproductive Study Ethics Committee of Peking University Third Hospital (2017SZ-035). Semen samples were selected through 40% density gradient of PureSperm (Nidacon International, Molndal, Sweden) by

centrifugation (500g, 30 min) at room temperature and washed with phosphate-buffered saline (PBS) for three times; the obtained spermatozoa were used for the eccDNA extraction.

Circle-Seq

Purification of mouse and human eccDNAs was performed as previously described, with minor modifications ((*Moller et al., 2018*)). In brief, samples were resuspended in 500 µl Lysis solution (10mM Tris pH7.4, 100mM NaCl, 1% SDS, 1% Sarkosyl, 150mM DTT) with 10µL Proteinase K (Thermo Scientific, EO0491), and incubated overnight at 55 °C. After cell lysis, phenol: chloroform: isoamyl alcohol was added and mixed, centrifuged at 13,000 g for 15min at 4°C. The supernatant was moved to a new tube, incubated with 500 µL isopropanol at room temperature for 10 min, and centrifuged at 13,000 g for 15min at 4 °C. The resulted DNA pellet was washed with 1 ml 70% ethanol, treated with alkaline to separate chromosomal DNA from eccDNAs by rapid DNA denaturing–renaturing, followed by column chromatography on an ion exchange membrane column (TIANprep Mini Plasmid Kit, DP103). Column-bound DNA was eluted in TE buffer (10 mM Tris-Cl, pH 8.0; 1 mM EDTA, pH 8.0). The remaining linear DNA was treated by exonuclease (Plasmid-Safe ATP-dependent DNase, Lucigen, E3101K) at 37°C for 1 week, during which additional ATP and DNase was added every 24 h (30 units per day) according to the manufacturer’s protocol (Plasmid-Safe ATP-dependent DNase, Lucigen, E3101K). The eccDNA samples were cleaned by phenol: chloroform: isoamyl alcohol once, followed by ethanol precipitation. EccDNAs were then amplified by phi29 polymerase (NEB, M0269L) at 30°C for 16 h. Phi29-amplified DNA was cleaned by phenol: chloroform: isoamyl alcohol once, followed by ethanol precipitation. The DNA samples were sonicated to a set size of 250 bp with an M220 Focused-ultra sonicator (Covaris, Woburn). Sequencing libraries were generated with NEBNext® Ultra™ II DNA Library Prep Kit for Illumina (NEB, E7645S), according to the manufacturer’s instructions. 10 ng of eccDNA samples were used for library preparation. NEBNext Multiplex Oligos for Illumina (Set 1, NEB #E7600) were used for PCR amplification of adaptor-ligated DNA. Libraries were purified with SPRIselect® Reagent Kit (Beckman Coulter, Inc #B23317). Paired-end 150 bp sequencing was performed on Illumina NovaSeq 6000 System.

EccDNA detection

We developed a nucleotide-resolution eccDNA detection pipeline on the basis of previous efforts ([Figure 4—figure supplement 1B](#)) ((*Kumar et al., 2017*); (*Moller et al., 2018*)). Briefly, SeqPrep (<https://github.com/jstjohn/SeqPrep>) was used to trim adapter sequences and merge the overlapping paired-end reads into singleton long reads, followed by reads mapping to GRCm38 reference genome using BWA MEM version 0.7.17-r1188 ((*Li et al., 2009*)). Samblaster version 0.1.26 ((*Faust et al., 2014*)) or an in-house Perl script was used to remove PCR duplicates and separate alignments into split reads, discordant and concordant reads. Candidate eccDNAs are firstly identified based on split reads (high-confidence ones). When homologous sequences are present at eccDNA ends, we recorded the coordinates of the leftmost form of eccDNA and an offset corresponding to the length of homologous sequences to represent all possible eccDNA variants. Potential split reads that failed to be mapped as split reads in the first place (low-confidence ones) as well as discordant reads were identified and counted using in-house Perl scripts. The average coverages (in terms of RPK) for candidate eccDNAs and surrounding regions were then calculated based on all different types of reads. Any eccDNA supported by at least two high-confidence split reads or discordant reads, with its 95% region covered by at least one read, and with its average coverage twice of that of its surrounding region, is considered as a high-confidence eccDNA.

EccDNA characterization

Gene structure annotations in mouse and human are based on Ensembl GRCh38 Release 102 and Ensembl GRCh38 Release 104, respectively. Repeat elements are annotated by RepeatMasker database. Random region generation, sequence composition calculation, and overlapping region determination were all done with Bedtools version 2.30.0 (*Quinlan et al., 2010*). The nucleosome occupancy probability of the eccDNAs and the surrounding regions was predicted using predNuPoP function of NuPoP version 2.2.0 R package (*Xi et al., 2010*). Motif analysis and visualization were done by Two Sample Logo web server (*Vacic et al., 2006*). Public ChIP-seq datasets (*Jung et al., 2019*); (*Jung et al., 2017*); (*Singh et al., 2021*) were aligned to GRCh38 reference genome using Bowtie, and sorted and indexed using SAMtools version 1.7 (*Danecek et al., 2021*). Picard MarkDuplicates version 2.18.14 was used to remove PCR duplicates. Peak calling was done with Macs2 version 2.2.7.1 (*Y. Zhang et al., 2008*). Coverage file was generated from BAM file and visualized using deeptools version 3.5.1 (*Ramirez et al., 2016*).

Data and software access

The raw sequencing data reported in this study have been deposited in the Genome Sequence Archive (*Chen et al., 2021*) in National Genomics Data Center (*Members et al., 2022*) (GSA accession numbers: CRA008015 and HRA002957) that are publicly accessible at <https://ngdc.cncb.ac.cn/gsa>. The eccDNA mapping and detection workflows are available at https://github.com/NjuChenlab/eccDNA_detector_tools.

Competing interest statement

The authors declare no competing financial interests.

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Author contributions

J.-Y.C., W.L., C.L. and H.J. conceived and designed the experiments. C.L. and Z.Z. performed the majority of the experiments with the help from S.X., Y.C., Y.C. and J.W., J.H. was responsible for the processing and analysis of most genomic data. J.-Y.C. interpreted the data and wrote the paper.

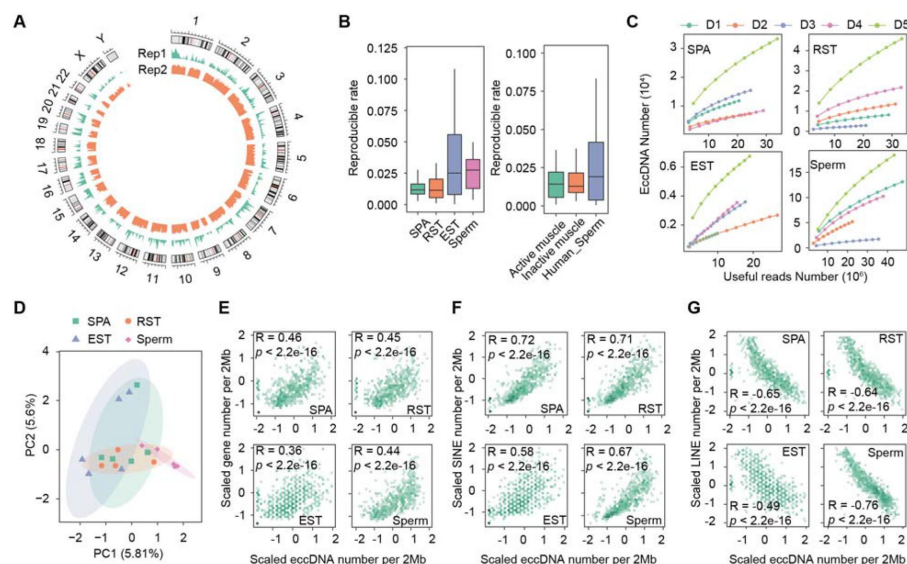


Figure 1—figure supplement 1.

Detection and characterization of eccDNAs. (A) Genome-wide distribution of eccDNAs detected in two human sperm samples. (B) Histograms showing distributions of reproducible rates between any two replicates of the same biological sample. Left: data generated in this study; Right: public datasets [Moller et al, Nat Commun (2018) 9:1069; Henriksen et al, Mol Cell (2022) 82: 209-217.e7]. (C) Number of detected eccDNAs as a function of non-redundant uniquely-mapped reads (useful reads) randomly sampled from indicated Circle-seq sequencing libraries.

(D) Principal component analysis of biological replicates using Jaccard indexes with all other replicates as features. 75% confidence ellipses are indicated. (E-G) eccDNA number per 2 mega bases as a function of gene density (E), density of SINE (F) and LINE (G) elements.

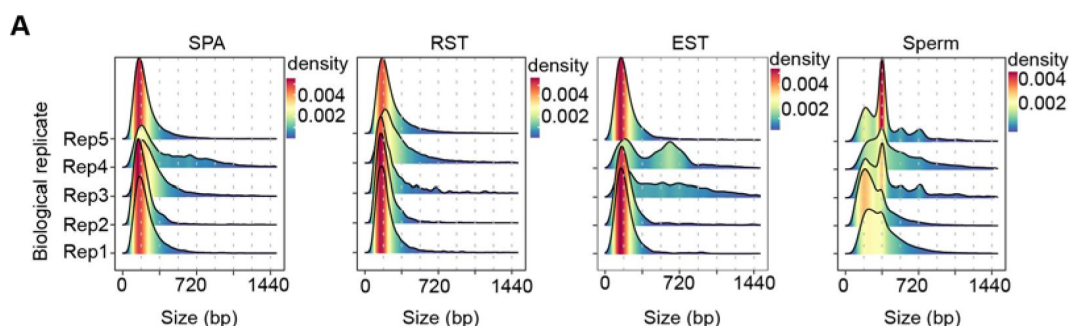


Figure 1—figure supplement 2.

Length distribution of eccDNAs in different cells. Dotted lines indicate multiplies of 180 bp.

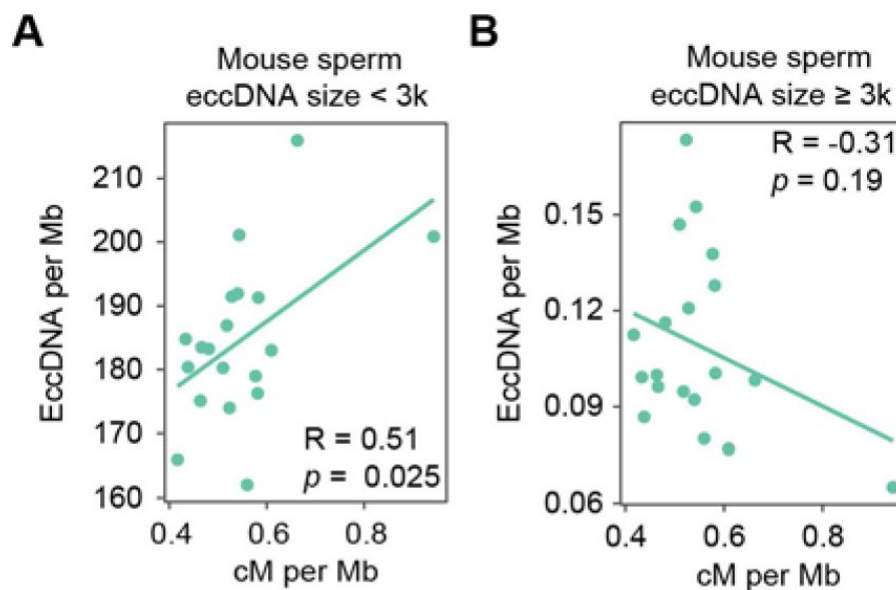


Figure 3—figure supplement 1.

Correlation between the density of small-sized (**A**) vs large-sized (**B**) eccDNAs and the meiotic recombination rate [Jensen-Seaman et al., *Genome Res* (2004) 14, 528-538].

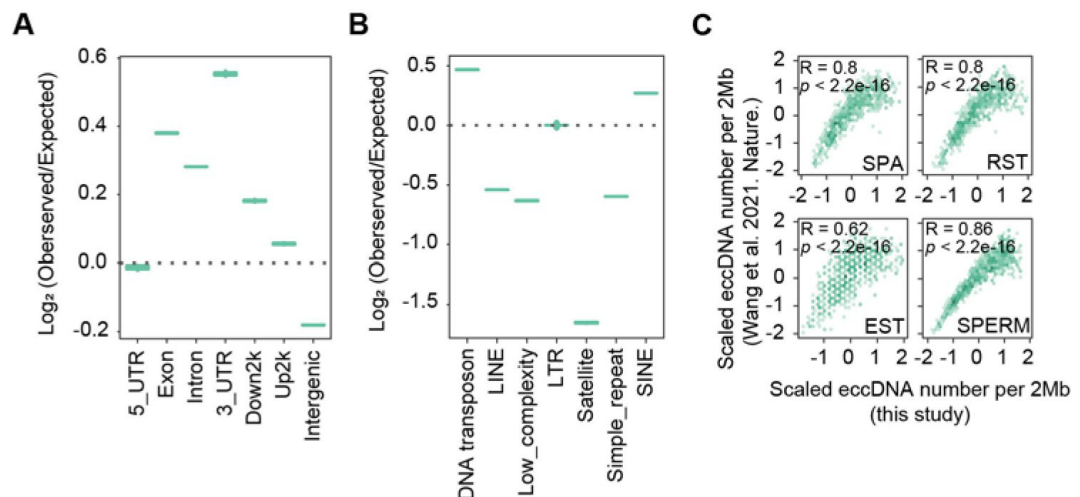


Figure 4—figure supplement 1.

High similarity between sperm eccDNAs detected in this study and those from apoptotic DNA fragmentation reported previously. (**A**) Enrichment of eccDNAs reported in Nature (2021) 599: 308-314 at given genomic regions relative to randomly-selected control regions. (**B**) Enrichment of eccDNAs reported in Nature (2021) 599: 308-314 at given repeat elements relative to randomly-selected control regions. (**C**) EccDNA numbers per 2Mb region in this study as a function of eccDNA numbers per 2 Mb reported in Nature (2021) 599: 308-314.

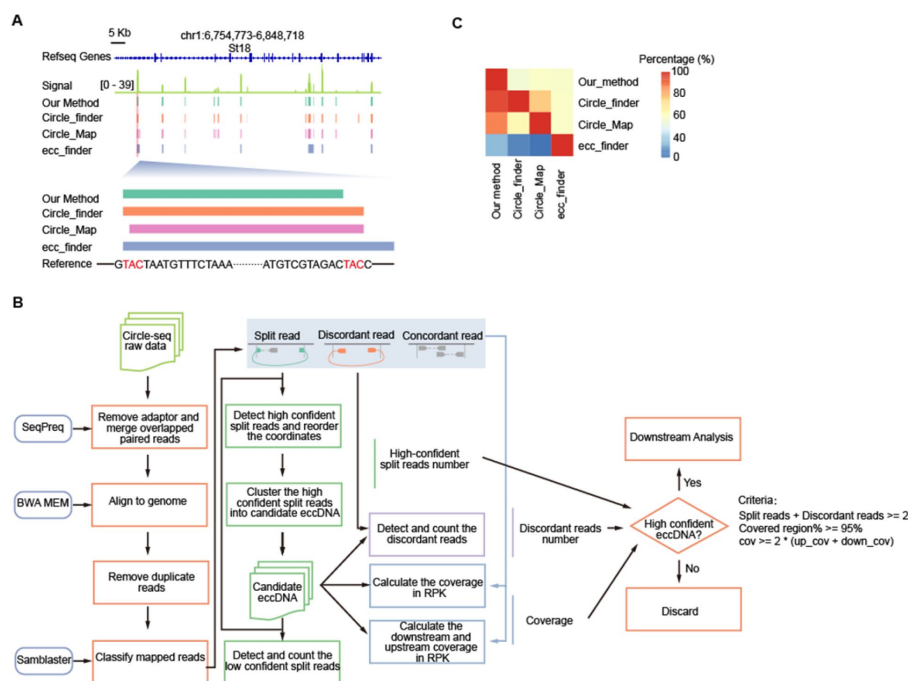


Figure 4—figure supplement 2.

Evaluation of our nucleotide-resolution eccDNA detection method. **(A)** A representative genomic locus showing Circle-seq signals and eccDNAs detected by individual methods. Enlarged are the first eccDNA as well as its sequence information. Homologous sequences (TAC) at the eccDNA end are highlighted in red. **(B)** Schematic representation of our eccDNA detection pipeline. **(C)** Fraction of eccDNAs detected by individual methods (x-axis) that are co-detected by other methods (y-axis).

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Reviewer #1 (Public Review):

This study aims to address the mechanism of eccDNA generation during spermatogenesis in mice. Previous efforts for cataloging eccDNA in mammalian germ cells have provided inconclusive results, particularly in the correlation between meiotic recombination and the generation of eccDNA. The authors employed an established approach (Circle-seq) to enrich and amplify eccDNA for sequencing analyses and reported that sperm eccDNA is not associated with meiotic recombination hotspots. Rather, the authors reported that eccDNAs are widespread, and oligonucleosomal DNA fragments from sperm undergoing apoptosis, with the ligation of DNA ends by microhomology-mediated end-joining, would be a major source of eccDNA.

The strength of the study includes evaluating the eccDNA contents not only in sperm but also from earlier stages of cells in spermatogenesis. The differences in eccDNA size peaks between sperm and other progenitors, in particular, the unique peak in sperm around 360 bp, are intriguing. Results from sequencing data analysis were presented elegantly.

I also have critiques. First, the lack of eccDNA quality control step is a concern. Previous studies employed electron microscopy to ensure that DNA species are mostly circular before rolling-circle amplification. Phi29 polymerase is widely used for DNA amplification, including whole genome amplification of linear chromosomal DNA. Phi29 polymerase has a high processivity and strand displacement activity. When those activities occur within a molecule, it creates circular DNA from linear DNA in vitro. In vitro-created eccDNA from linear DNA would be randomly distributed in the genome, which may explain the low incidence of common eccDNA between replicates. Therefore, it will be crucial to show that DNA prior to amplification is dominantly circular. Electron microscopy would be challenging for the study because the relatively small number of cells were processed to enrich eccDNA. An alternative method for quality controls includes spiking samples with linear and circular exogenous DNA and measuring the ratios of circular/linear control DNA before and after column purification/exonuclease digestion. eccDNA isolation procedures can be validated by a very high circular/linear control DNA ratio.

Another critique is regarding the limitation of the study. It is important to remind the readers of the limitations of the study. As the authors mentioned, rolling circle amplification preferentially increases the copy numbers of smaller eccDNA. Therefore, the native composition of eccDNA is skewed. In addition, the candidate eccDNAs are identified by split reads or discordant read pairs. The details of the mapping process are unclear from the methods, but such a method would require reads with high mapping quality; the identification of eccDNA is expected to require sequencing reads that are mapped to genomic locations uniquely with high confidence, and reads mapped to more than one genomic location, such as highly similar repeat sequences or duplications, are eliminated. Such identification criteria would favor eccDNA formed by little or no homology at the junction sequences, and eliminate eccDNA formed by long homologies at the ends, such as eccDNA formed exclusively by satellite DNA. Therefore, it is not surprising that the authors found the dominance of microhomology-mediated eccDNA. It remains to be determined whether small eccDNA with microhomologies are the dominant species of eccDNA in the native composition. In this regard, it is noted that similar procedures of eccDNA enrichment (column purification, exonuclease digestion, and rolling circle amplification) revealed variable sizes and characteristics of eccDNA in sperm (human from Henriksen et al. or mice from this study), dependent on the methods of sequencing (long-read or short-read sequencing). Considering these limitations, the last sentence of the introduction, "We conclude that germline eccDNAs are formed largely by microhomology mediated ligation of nucleosome protected fragments, and barely contribute to de novo genomic deletions at meiotic recombination hotspots" needs to be revised.

Small eccDNA (microDNA) data from various mouse tissues are available from the study by Dillion et al., (Cell Reports 2015). Authors are encouraged to examine whether the notable findings in this study (oligonucleosomal-sized eccDNA peaks and the association with apoptotic cell death) are unique to sperm or common in the eccDNA from other tissues.

Reviewer #2 (Public Review):

This study presents a useful investigation of eccDNAs in spermatogenesis of mouse. It provides evidence about the biogenesis of eccDNAs and suggests that eccDNAs are derived from oligonucleosomal DNA fragmentation during apoptosis by MMEJ and may not be the direct products of germline deletions. However, the method of data analyses were not fully described and data analysis is incomplete. It provides additional observations about the eccDNA biogenesis and can be used as a starting point for functional studies of eccDNA in sperms. However, many aspects about data analyses and data interpretations need to be improved.

- Most of the conclusions made by the work are only based on the bioinformatics analyses, the validation of these findings using other method (biochemistry/molecular biology method) are missing. For example, no QC results presented for the eccDNA purification, which may show whether contaminants such as linear DNA or mitochondria DNA have been fully removed. Additionally, it is also helpful to use simple PCR to test the existence of identified eccDNAs in sperm or other samples to validate the specificity of the Circle-seq method.
- The reliability of the data analysis methods is uncertain, as the authors constructed and utilized their own pipeline to identify eccDNAs, despite the availability of established bioinformatics tools such as ECCplorer, eccFinder, and Amplicon Architect. Moreover, the lack of validation of the pipeline using either ground truth datasets or simulation data raises concerns about its accuracy. Additionally, the methodology employed for identifying eccDNA that encompasses multiple gene loci remains unclear.

- Although the author stated that previous studies utilizing short-read sequencing technologies may have incorrectly annotated eccDNA breakpoints, this claim requires careful scrutiny and supporting evidence, which was not provided in the manuscript.
- The similarity between the eccDNA profiles of human and mouse sperm remains uncertain, and therefore, analyses of human eccDNA data and comparisons between the two are necessary if the authors claim that their findings of widespread eccDNA formation in mouse spermatogenesis extend to human sperms.

Author Response

eLife assessment

This study provides valuable information on the biogenesis of eccDNAs during spermatogenesis, i.e., eccDNAs in spermatogenic cells are not derived from mitotic recombination hotspots but represent oligonucleosomal DNA fragments from apoptotic male germ cells, whose ends are ligated through microhomology-mediated end-joining. The study is currently incomplete because the method of bioinformatics needs more details and data interpretation should take the amplification bias into consideration.

We highly appreciate the positive assessment.

The negative assessment of our bioinformatics method is probably based on Reviewer #2's comments. While Reviewer #1 considered that "Results from sequencing data analysis were presented elegantly", Reviewer #2 overlooked some details and raised several critiques regarding our bioinformatics method. We respectfully disagree with many of his or her critiques: (I) Reviewer #2 considered that our method was not fully described. However, we have illustrated the principle and steps of our eccDNA detection method by Figure 4C and Figure 4-figure supplement 2, and submitted our source codes to GitHub. (II) Reviewer #2 had concerns on the reliability of our method. However, we have revealed that it has comparable sensitivity and specificity with established bioinformatics tools (Figure 4—figure supplement 2C), and even higher accuracy on the assignment of eccDNA boundaries (Figure 4—figure supplement 2A). (III) Reviewer #2 also believed that "the similarity between the eccDNA profiles of human and mouse sperm remains uncertain". However, we believe that our Fig. 5 have clearly shown that human sperm eccDNAs have exactly the same characteristics with mouse sperm eccDNAs. Nevertheless, in revised manuscript, we will add more description to help readers to better understand our method, and perform additional analyses to further back up our claims.

The amplification bias is indeed a problem of Circle-seq. Following editors' and Reviewer #1's insightful suggestions, we will analyze other datasets generated either by rolling circle amplification or not to see how our findings are affected. Additionally, we will consider to add one section to remind readers of the limitations of rolling-circle amplification-based Circle-seq and our data interpretation.

Reviewer #1 (Public Review):

This study aims to address the mechanism of eccDNA generation during spermatogenesis in mice. Previous efforts for cataloging eccDNA in mammalian germ cells have provided inconclusive results, particularly in the correlation between meiotic recombination and the generation of eccDNA. The authors employed an established approach (Circle-seq) to enrich and amplify eccDNA for sequencing analyses and reported that sperm eccDNA is not associated with mitotic recombination hotspots. Rather, the authors reported that eccDNAs are widespread, and oligonucleosomal DNA fragments from sperm undergoing apoptosis, with the ligation of DNA ends by microhomology-mediated end-joining, would be a major source of eccDNA.

The strength of the study includes evaluating the eccDNA contents not only in sperm but also from earlier stages of cells in spermatogenesis. The differences in eccDNA size peaks between sperm and other progenitors, in particular, the unique peak in sperm around 360 bp, are intriguing. Results from sequencing data analysis were presented elegantly.

We are grateful to Reviewer #1 for his or her recognition of the strength of this study.

I also have critiques. First, the lack of eccDNA quality control step is a concern. Previous studies employed electron microscopy to ensure that DNA species are mostly circular before rolling-circle amplification. Phi29 polymerase is widely used for DNA amplification, including whole genome amplification of linear chromosomal DNA. Phi29 polymerase has a high processivity and strand displacement activity. When those activities occur within a molecule, it creates circular DNA from linear DNA in vitro. In vitro-created eccDNA from linear DNA would be randomly distributed in the genome, which may explain the low incidence of common eccDNA between replicates. Therefore, it will be crucial to show that DNA prior to amplification is dominantly circular. Electron microscopy would be challenging for the study because the relatively small number of cells were processed to enrich eccDNA. An alternative method for quality controls includes spiking samples with linear and circular exogenous DNA and measuring the ratios of circular/linear control DNA before and after column purification/exonuclease digestion. eccDNA isolation procedures can be validated by a very high circular/linear control DNA ratio.

We highly appreciate Reviewer #1's insightful suggestions. We would like to perform eccDNA quality control by introducing circular exogenous DNA into our samples and measuring its ratio to endogenous linear DNA before and after eccDNA isolation procedures.

Another critique is regarding the limitation of the study. It is important to remind the readers of the limitations of the study. As the authors mentioned, rolling circle amplification preferentially increases the copy numbers of smaller eccDNA. Therefore, the native composition of eccDNA is skewed. In addition, the candidate eccDNAs are identified by split reads or discordant read pairs. The details of the mapping process are unclear from the methods, but such a method would require reads with high mapping quality; the identification of eccDNA is expected to require sequencing reads that are mapped to genomic locations uniquely with high confidence, and reads mapped to more than one genomic location, such as highly similar repeat sequences or duplications, are eliminated. Such identification criteria would favor eccDNA formed by little or no homology at the junction sequences, and eliminate eccDNA formed by long homologies at the ends, such as eccDNA formed exclusively by satellite DNA. Therefore, it is not surprising that the authors found the dominance of microhomology-mediated eccDNA. It remains to be determined whether small eccDNA with microhomologies are the dominant species of eccDNA in the native composition. In this regard, it is noted that similar procedures of eccDNA enrichment (column purification, exonuclease digestion, and rolling circle amplification) revealed variable sizes and characteristics of eccDNA in sperm (human from Henriksen et al. or mice from this study), dependent on the methods of sequencing (long-read or short-read sequencing). Considering these limitations, the last sentence of the introduction, "We conclude that germline eccDNAs are formed largely by microhomology mediated ligation of nucleosome protected fragments, and barely contribute to de novo genomic deletions at meiotic recombination hotspots" needs to be revised.

We thank Reviewer #1 for pointing out limitations of the study. We will take into account and integrate the perspectives of Reviewer #1 in our revised manuscript. We will also try to analyze eccDNA datasets generated by long-read sequencing to see how our conclusions might be affected. However, we envision that it might be challenging to examine the contribution of microhomology-mediated ligation to eccDNA biogenesis using long-read sequencing data as the sequencing error rate of nanopore long-read sequencing data is very high.

Small eccDNA (microDNA) data from various mouse tissues are available from the study by Dillion et al., (Cell Reports 2015). Authors are encouraged to examine whether the notable findings in this study (oligonucleosomal-sized eccDNA peaks and the association with apoptotic cell death) are unique to sperm or common in the eccDNA from other tissues.

We are thankful to Reviewer #1 for this suggestion. We would like to analyze additional eccDNA sequencing datasets to see whether our findings are unique to sperm or common for other tissues.

Reviewer #2 (Public Review):

This study presents a useful investigation of eccDNAs in spermatogenesis of mouse. It provides evidence about the biogenesis of eccDNAs and suggests that eccDNAs are derived from oligonucleosomal DNA fragmentation during apoptosis by MMEJ and may not be the direct products of germline deletions. However, the method of data analyses were not fully described and data analysis is incomplete. It provides additional observations about the eccDNA biogenesis and can be used as a starting point for functional studies of eccDNA in sperms. However, many aspects about data analyses and data interpretations need to be improved.

We thank Reviewer #2 for his or her critical reading. However, we respectfully disagree with some critiques on our data analyses (see below). Anyway, we will provide more method details in addition to Fig. 4C and Figure 4-figure supplement 2 that have illustrated the principle and steps of our method, as well as the performance in comparison with established methods. We will also perform additional analyses and make some clarifications in revised manuscript (see below).

- *Most of the conclusions made by the work are only based on the bioinformatics analyses, the validation of these findings using other method (biochemistry/molecular biology method) are missing. For example, no QC results presented for the eccDNA purification, which may show whether contaminants such as linear DNA or mitochondria DNA have been fully removed. Additionally, it is also helpful to use simple PCR to test the existence of identified eccDNAs in sperm or other samples to validate the specificity of the Circle-seq method.*

Following both this Reviewer's and Reviewer #1's suggestions, we will introduce circular exogenous DNA into our samples and measure its ratio to endogenous linear DNA and to mitochondria DNA before and after eccDNA isolation procedures. We will also try to perform PCR to test the existence of identified eccDNAs.

- *The reliability of the data analysis methods is uncertain, as the authors constructed and utilized their own pipeline to identify eccDNAs, despite the availability of established bioinformatics tools such as ECCsplorer, eccFinder, and Amplicon Architect. Moreover, the lack of validation of the pipeline using either ground truth datasets or simulation data raises concerns about its accuracy. Additionally, the methodology employed for identifying eccDNA that encompasses multiple gene loci remains unclear.*

In fact, we have compared the performance between our method and established methods for identification of eccDNA regions, such as Circle_finder, Circle_Map and ecc_finder. Our method has comparable sensitivity and specificity with existing methods, especially Circle_finder and Circle_Map (Figure 4—figure supplement 2C). We also used one specific genomic region to show that existing methods identified the same eccDNA regions but misassigned the eccDNA boundaries (Figure 4—figure supplement 2A). These results have been shown in Figure 4—figure supplement 2. We will highlight the information to make it more clear in our revised manuscript. We will further detect eccDNAs by ECCsplorer for comparison. Since Amplicon Architect is more specifically designed for detection of ecDNAs, it will not be included in our comparison. We will also try to perform PCR to validate the identified eccDNAs.

As pointed out by Reviewer #2, similar to ECCsplorer, Circle_finder, Circle_Map and ecc_finder, our method fails to identify ecDNAs that encompass multiple gene loci. We will remind readers of this limitation in our revised manuscript.

- *Although the author stated that previous studies utilizing short-read sequencing technologies may have incorrectly annotated eccDNA breakpoints, this claim requires careful scrutiny and supporting evidence, which was not provided in the manuscript.*

As abovementioned, we used one specific genomic region to show that existing methods all misassigned the eccDNA boundaries (Figure 4—figure supplement 2A). In revised manuscript, we will provide necessary statistics to support this claim.

- *The similarity between the eccDNA profiles of human and mouse sperm remains uncertain, and therefore, analyses of human eccDNA data and comparisons between the two are necessary if the authors claim that their findings of widespread eccDNA formation in mouse spermatogenesis extend to human sperms.*

We believe that our Fig. 5 have clearly shown that human sperm eccDNAs are also originated from oligonucleosomal fragmentation (Fig. 5A-C), not associated with meiotic recombination hotspots (Fig. 5D and E) but formed by microhomology directed ligation (Fig. 5F and G). These findings are consistent with what we observed in mouse sperm eccDNAs. Nevertheless, we will analyze additional public datasets to further back up our claim in revised manuscript.